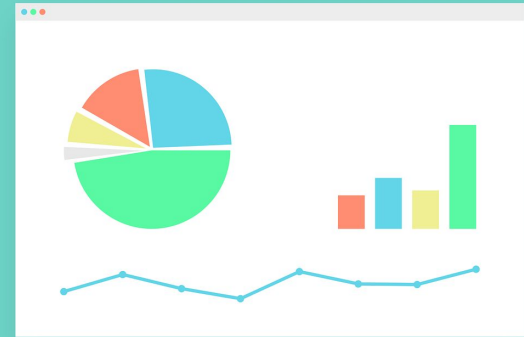


Lesson 16: Graphics

HS192 Introduction to Writing - 2
@ IITGN, 2025-26



Today's learning objective- 1

Analyze and create graphically represented data

Have you seen graphics in academic writing?

What types of graphics have you seen in textbooks and academic articles?

Why do we use graphics for academic purposes?

We use graphics to ...

Enhance the reader's understanding of the information in the document. Graphics are categorized into two:

- **Tables:** graphics that use rows and columns to structure & organise information
- **Figures:** anything other than a table, s.a. images and transcripts.

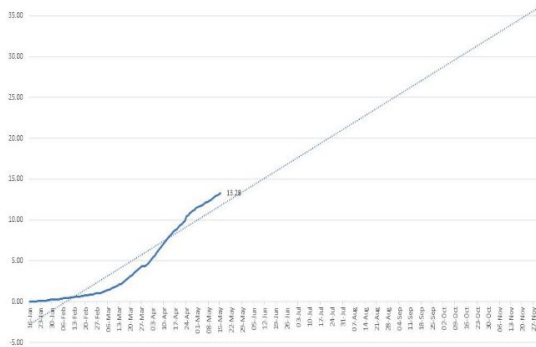


Figure 1: Linear projection of the population that will be vaccinated at the current pace of vaccination.

Table 1: Comparison of cases and deaths in both the COVID-19 waves in the highest 25 districts.¹⁴

Comparison	1st COVID wave (24 March to 1 October 2020)	2nd COVID wave (1 March to 15 May 2021)
Cumulative cases in India	6391480	13558915
Cumulative cases of 25 districts	2481248	4881679
% of cases contributed by these 25 districts	38.8	36.0
Cumulative deaths in India	99806	113033
Cumulative deaths in 25 districts	47667	43037
% of deaths contributed by these 25 districts	47.8	38.1

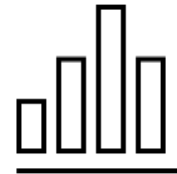
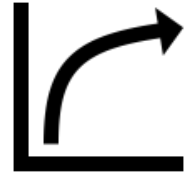
https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_formatting_and_style_guide/apa_tables_and_figures.html

Charts, Tables, Graphs....

Are useful to **summarize** data.

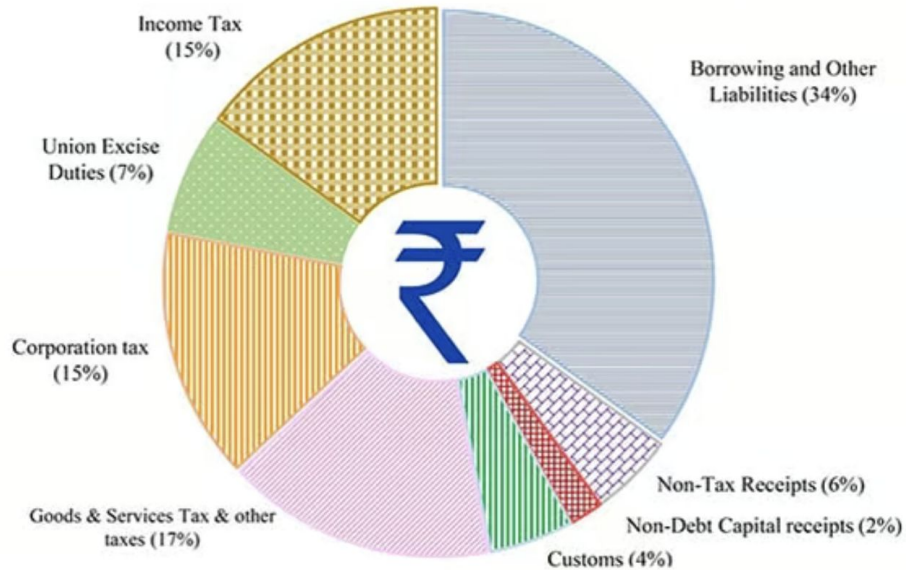
Show **trends** across data (a trend is a directional change in data).

Help to see the amount of **variation** in the information.

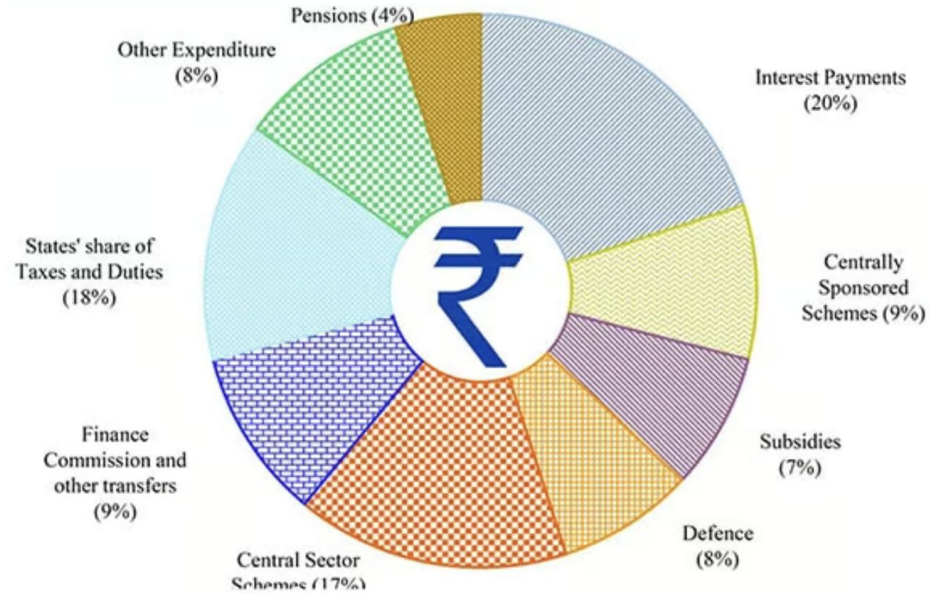


Pie chart

Rupee Comes From



Rupee Goes To



(Source: ["Explained: How India Earns Each Rupee And Where It Goes"](#). NDTV Profit. Accessed: Feb. 27, 2023)

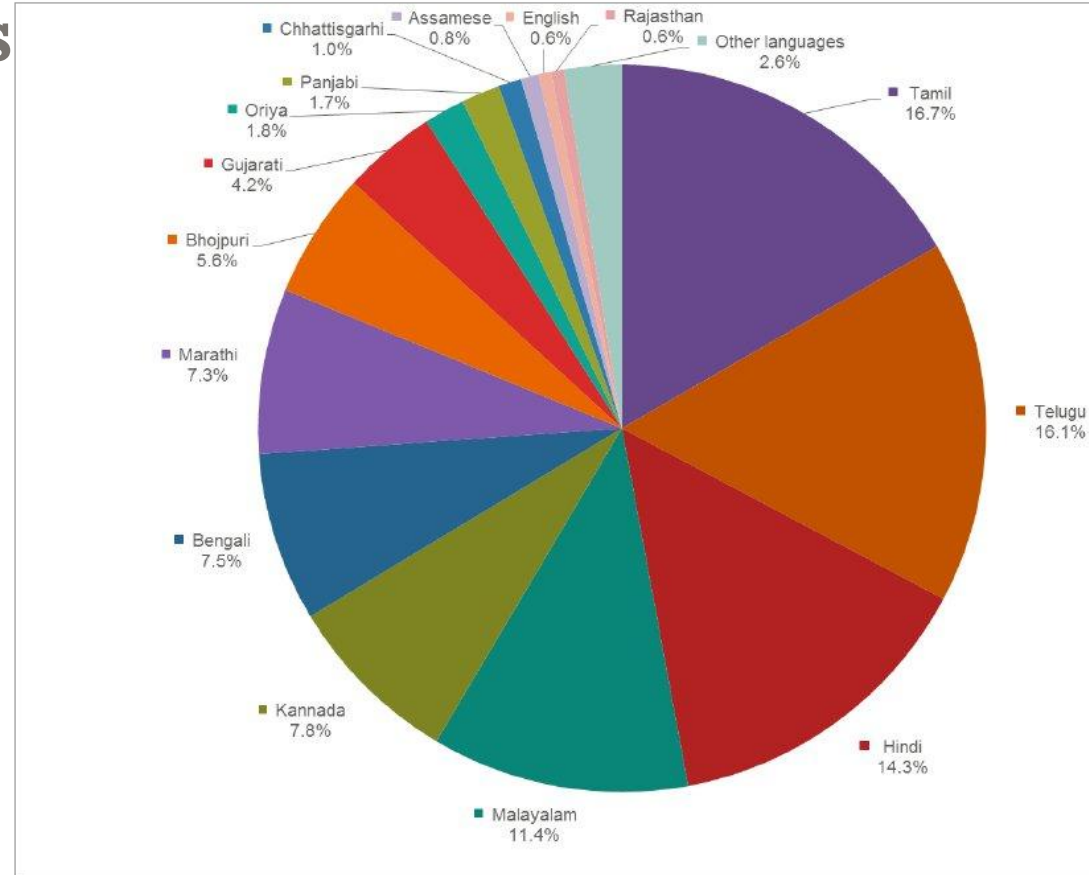
Pie chart components

Topic (caption)

Value

Legend

Figure 5. Languages used in feature films in India, 2012-2013



Source: Based on Film Federation of India data (2012 and 2013)

(Source: [UNESCO Institute for Statistics](#), May 21, 2016.
Accessed: Feb. 27, 2024)

Bar graph

Topic (caption)

X axis

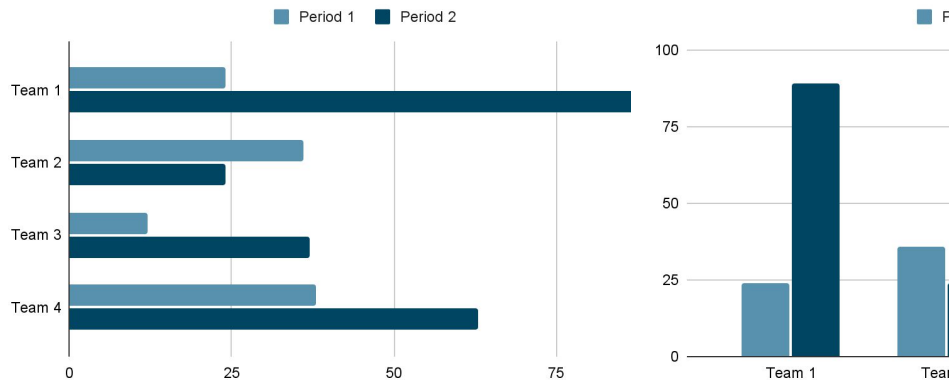
Y axis

Value/unit

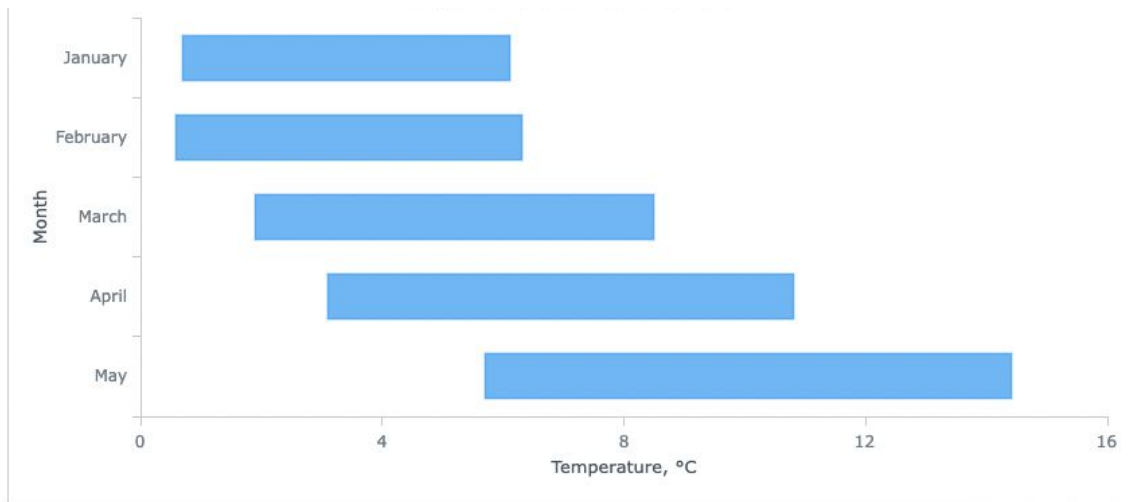
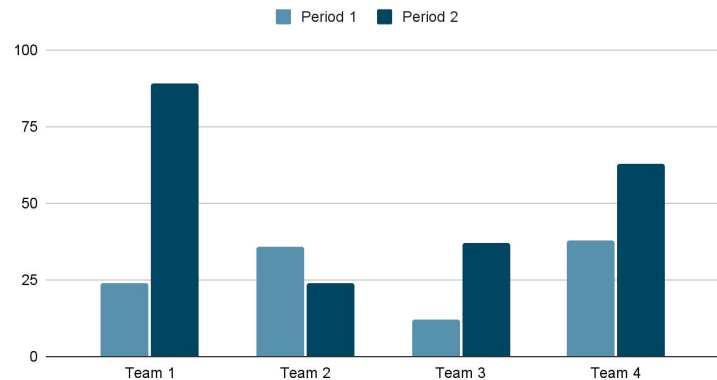
Trends

Range

Points scored



Points scored



Bar graph

Topic

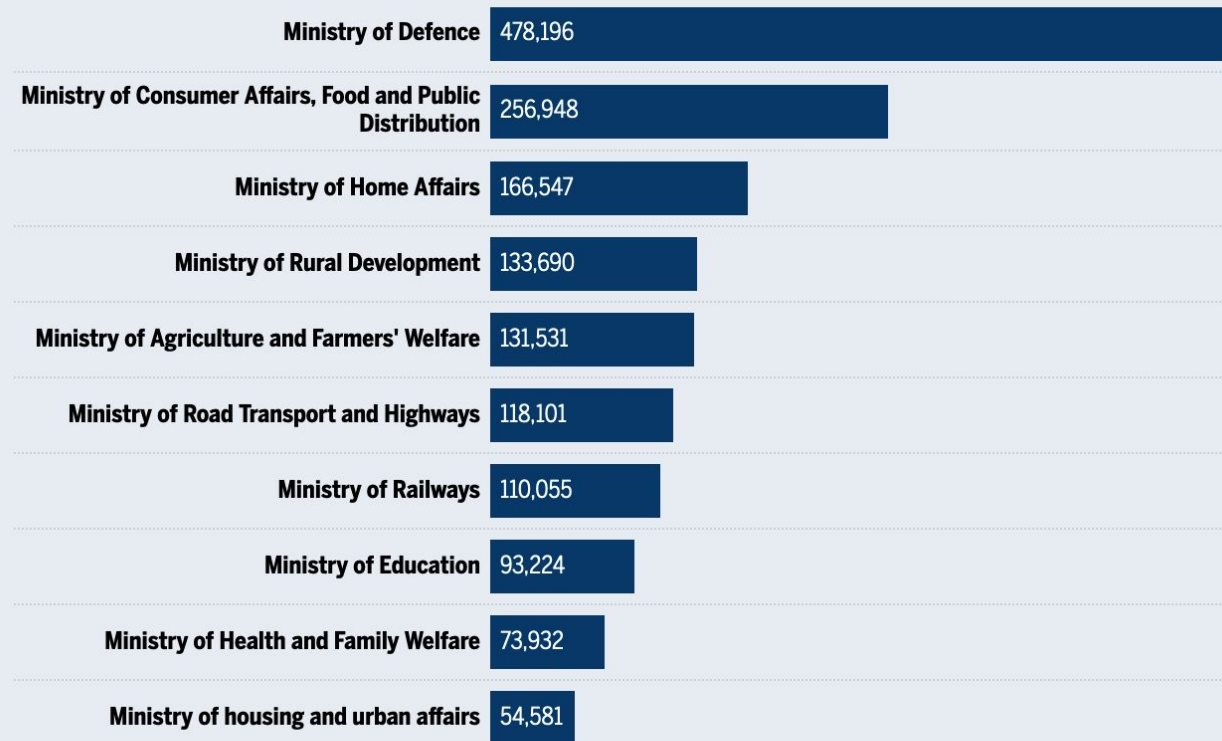
X axis

Y axis

Value/unit

Major Allocations

(in Rs crore)



TOI

Source: Government of India

Line graph

Topic

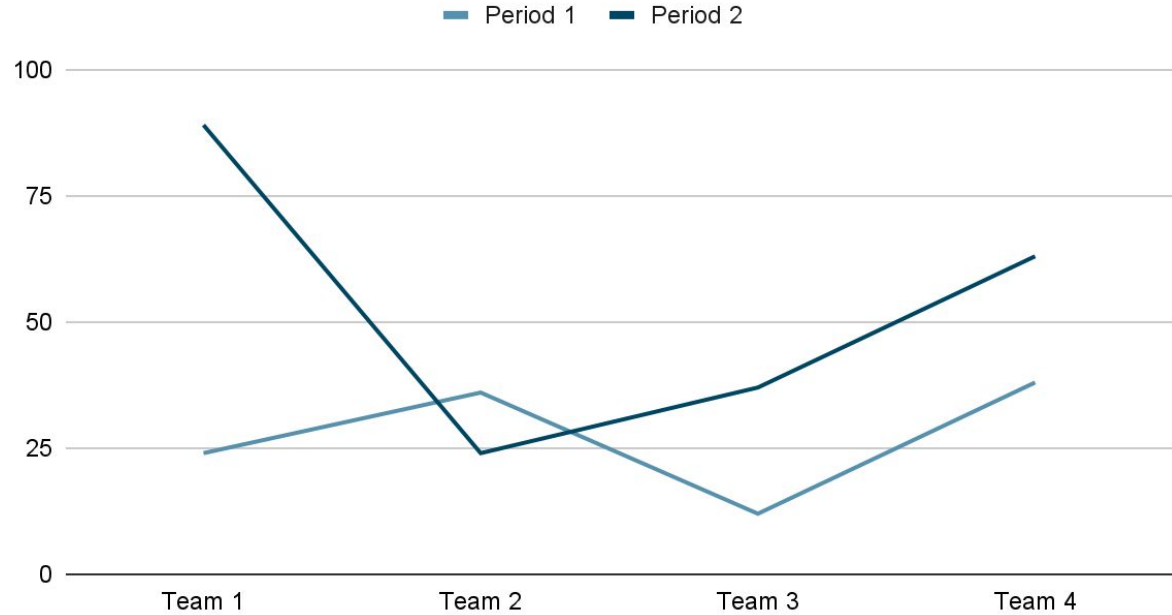
X axis

Y axis

Value/unit

Trend

Points scored



Line graph

Topic

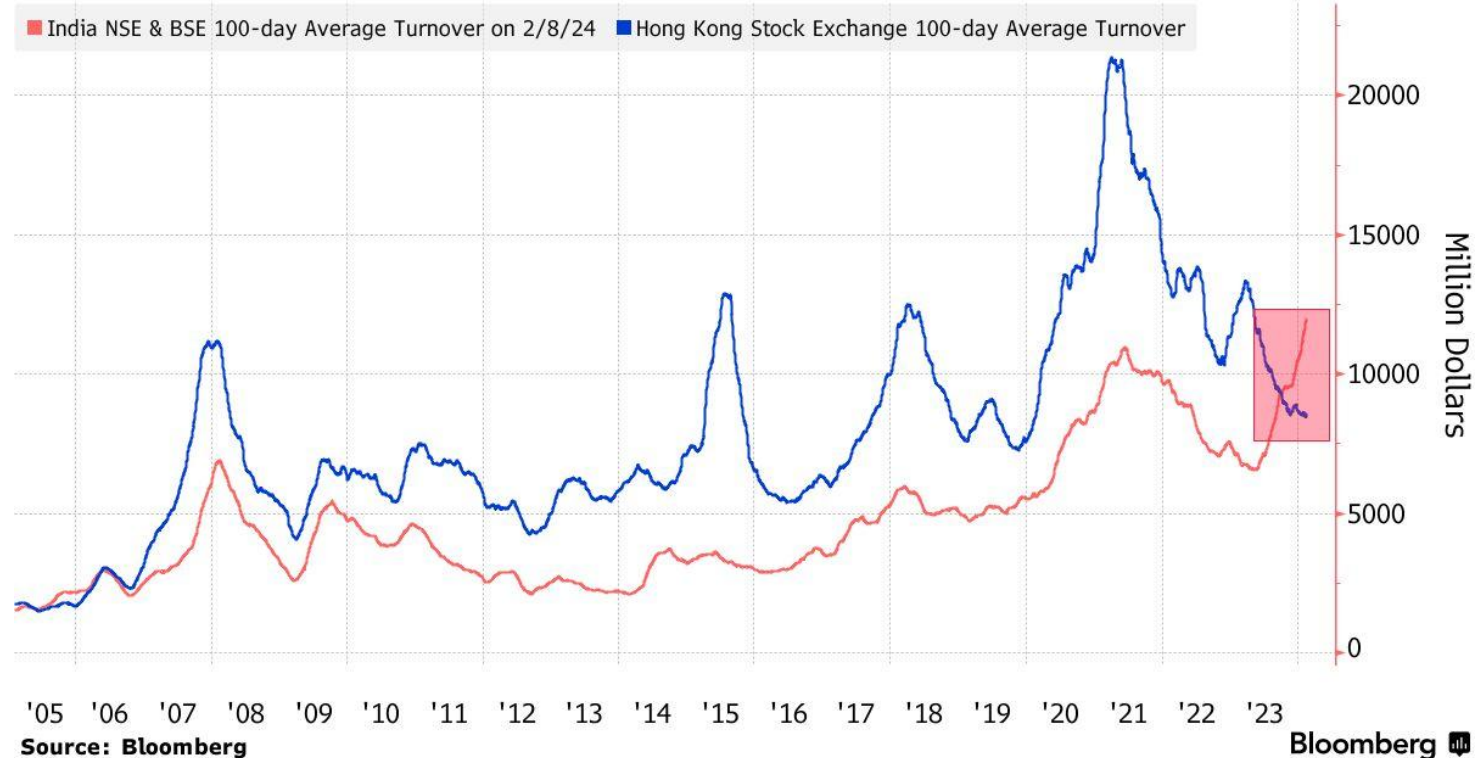
X axis

Y axis

Value

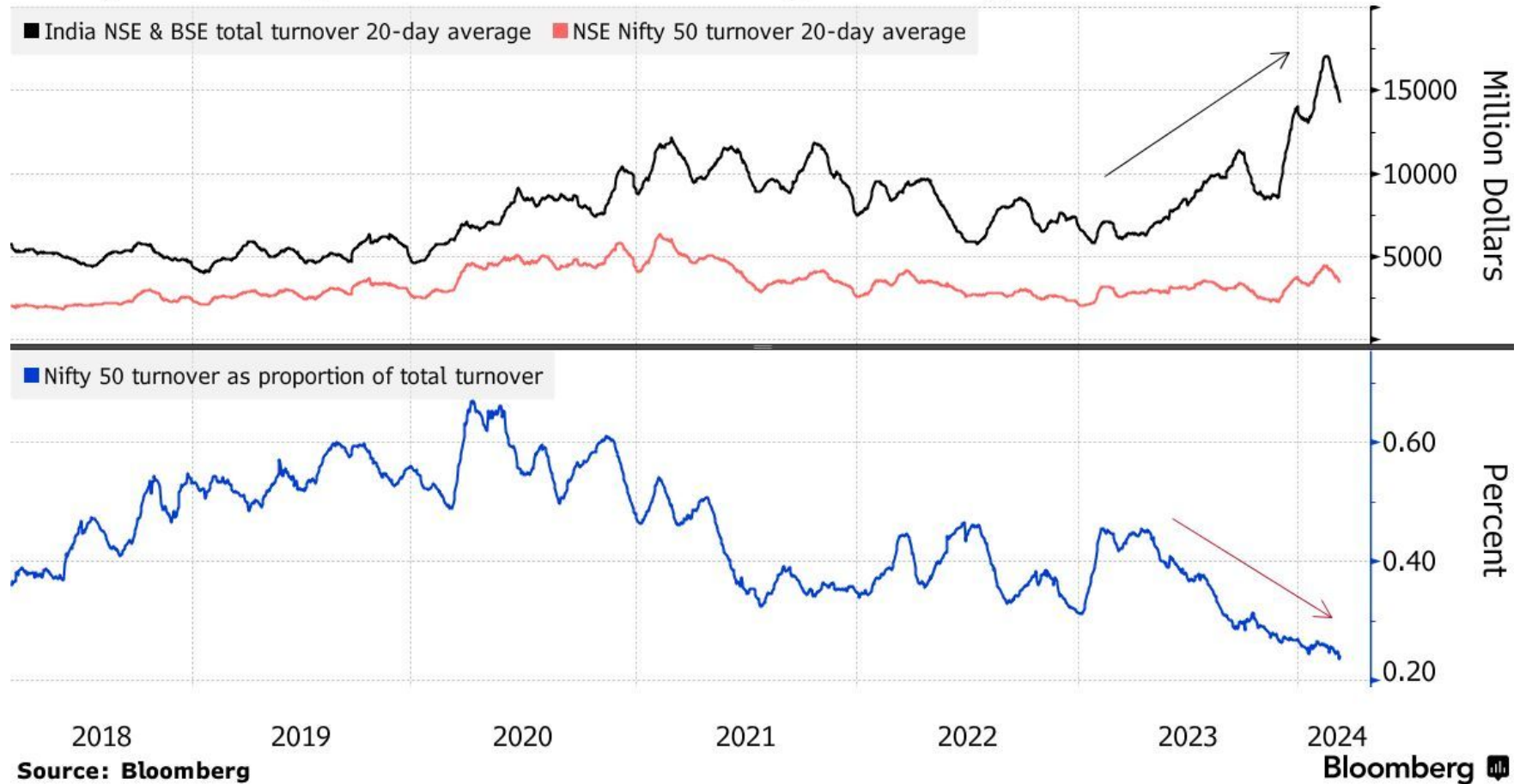
Trend

Daily Trading Value in India Overtakes Hong Kong



(Source: <https://x.com/business/status/1755860846339629292?s=20>)

Daily Trading Value in India Surge Led by Broader Market



(Source: <https://www.bloomberg.com/news/articles/2024-03-06/global-funds-catch-craze-for-india-s-minnows-in-hunt-for-winners>)

Table

Table 1: Developmental Capital Expenditure in India
(Amount in crore rupees)

Topic	Year	Developmental Expenditure	% of TE	% of GDP
Row	1990	16502	10.8	3.1
Column	1995	23767	8.3	2.1
	2000	41140	7.6	2.1
	2005	97566	10.4	2.9
	2010	213662	10.1	2.9
	2015	470635	12.4	3.8
	CAGR	14.3%	0.6%	0.8%

Source: Indian Public Finance Statistics, Ministry of Finance, Govt. of India (related issues)

Table: Find the trends of international currencies / USD

Country	Currency	2023	2022	2021	2020	2019	2018
Afghanistan	Afghani	82.635	90.08	83.48	76.65	77.58	73.60
Algeria	Dinar	135.933	142.12	135.01	126.74	119.40	117.41
Argentina	Peso	296.154	130.79	95.10	70.64	48.19	28.17
Australia	Dollar	1.506	1.44	1.33	1.45	1.44	1.34
Bahrain	Dinar	0.377	0.38	0.38	0.38	0.38	0.40
Brazil	Real	4.994	5.17	5.40	5.15	3.95	3.66
Canada	Dollar	1.35	1.30	1.25	1.34	1.33	1.30
Cayman Islands	Dollar	0.833	0.83	0.83	0.83	0.83	0.83
China	Yuan	7.075	6.73	6.45	6.90	6.91	6.62
Denmark	Krone	6.89	7.08	6.29	6.54	6.67	6.32
Egypt	Pound	30.651	19.21	15.70	15.81	16.81	17.81
Euro Zone	Euro	0.924	0.95	0.85	0.88	0.89	0.85
Hong Kong	Dollar	7.829	7.83	7.77	7.76	7.84	7.84
Hungary	Forint	353.02	372.78	303.29	307.77	290.71	270.44
Iceland	Krona	137.857	135.30	126.99	135.35	122.57	116.38
India	Rupee	82.572	78.60	73.94	74.10	70.39	68.42
Iraq	Dinar	1376.529	1,459.75	1,460.13	1,197.50	1,191.25	1,193.48
Israel	New Shekel	3.687	3.36	3.23	3.44	3.56	3.60
Japan	Yen	140.511	131.45	109.82	106.73	109.01	110.42
Lebanon	Pound	1370.988	1,515.67	1,519.23	1,510.68	1,510.29	1,511.68
Mexico	Peso	17.733	20.11	20.28	21.47	19.25	19.23
Morocco	Dirham	10.134	10.28	9.00	9.50	9.61	9.39
New Zealand	Dollar	1.63	1.58	1.42	1.54	1.52	1.45
Norway	Kroner	10.564	9.62	8.60	9.41	8.80	8.14
Qatar	Rial	3.643	3.64	3.64	3.64	3.64	3.64
Russia	Ruble	85.509	69.90	73.69	72.30	64.69	62.85
Saudi Arabia	Riyal	3.752	3.76	3.75	3.75	3.75	3.75
Singapore	Dollar	1.343	1.38	1.34	1.38	1.36	1.35
South Africa	Rand	18.457	16.38	14.79	16.46	14.45	13.26
South Korean	Won	1306.686	1,291.73	1,144.88	1,179.20	1,165.70	1,100.59
Sweden	Krona	10.613	10.12	8.58	9.21	9.46	8.70
Switzerland	Franc	0.899	0.96	0.91	0.94	0.99	0.98
Taiwan	Dollar	31.16	29.81	27.93	29.46	30.90	30.15
Thailand	Baht	34.802	35.04	32.00	31.27	31.03	32.32
Tunisia	Dinar	3.103	3.08	2.78	2.84	2.93	2.71
Turkey	New Lira	23.824	16.57	8.90	7.03	5.69	4.85
United Arab Emirates	Dirham	3.673	3.67	3.67	3.67	3.67	3.67
United Kingdom	Pound	0.804	0.81	0.73	0.78	0.78	0.75
Venezuela	Bolivar (Fuerte)	2,863,377,461,538.50	666,470,505,836.60	323,298,866,894.80	236,266.51	248,486.04	131,352.21

(Source:
<https://universityoftaxprofessionals.com/2023-irs-exchange-rates/>)

Table 2: 2018-2023 Average IRS Exchange Rates for Converting Foreign Currencies into US Dollars

Summary

Graphical representation helps communicate large amounts of information efficiently.

Graphical representation can be figures (pie charts, bar graphs etc) or tables.

They can summarize data & make it easy to see trends.

Trends are a direction of change in the data.

The variation of data available for the information is also very visible with graphical representation

Group activity

Directions:

Read the table/figures on the next slides.

Answer the questions together.

Come up with one answer for all questions.

Q1.

Were there more contributions by faculty or post graduates?

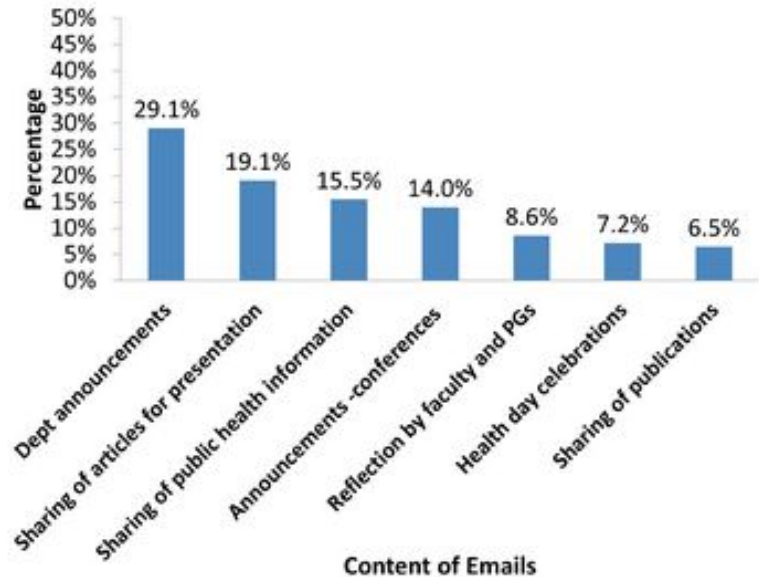
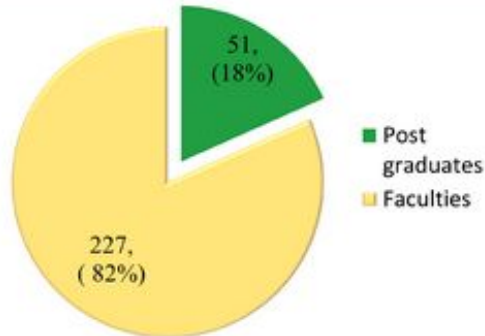
Q2.

Which type of email had the highest amount of presence?

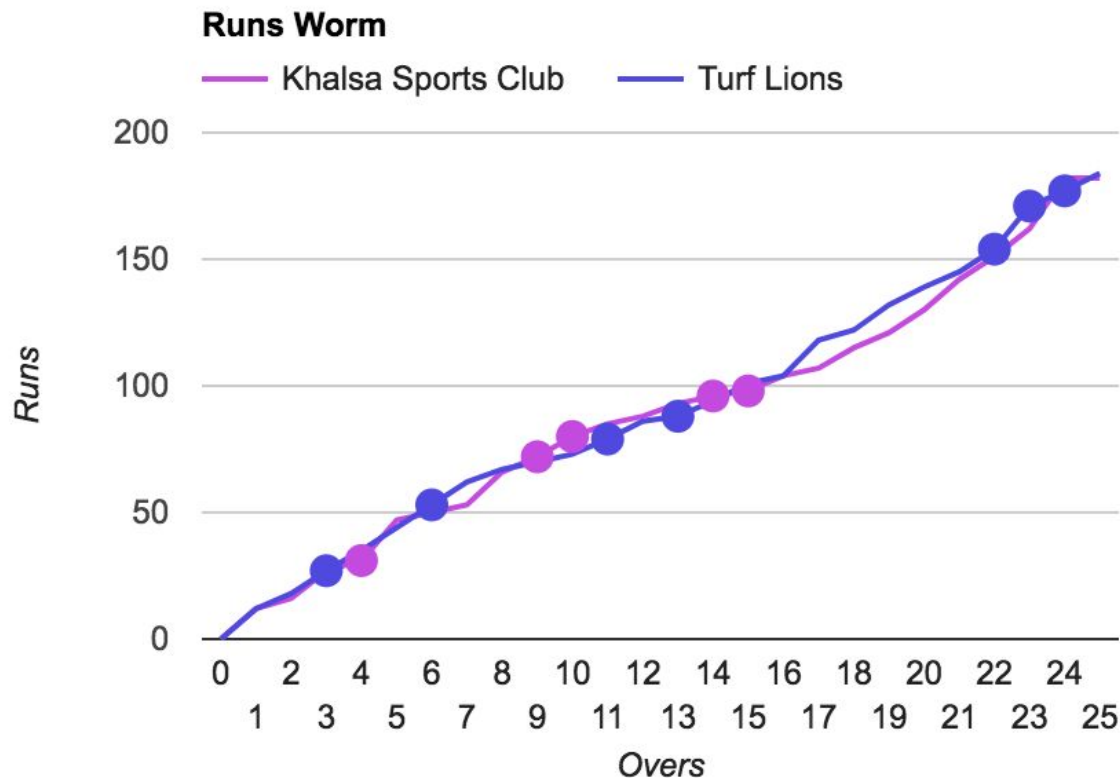
Q3.

Which all type of email content were below 15% of the total emails sent?

Distribution of Contribution by Post graduates and faculties



Groups 1-4: task 1/2



Q1.

At Over 11, which team had scored higher number of runs?

Q2.

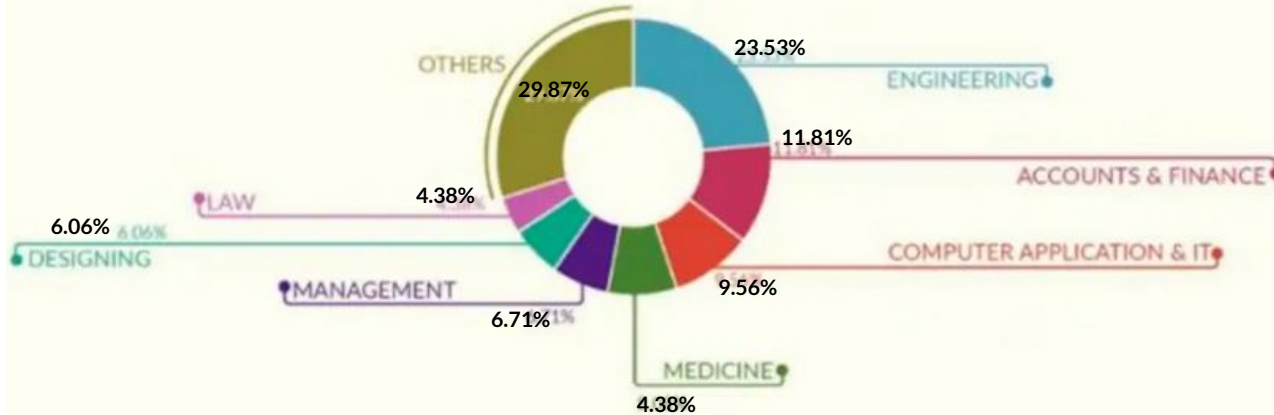
Name 2 overs when the 2 teams had the same score.

Q3.

At over 20, which team was more likely to score more runs?

Groups 1-4: task 2/2

COMMON CAREER OPTIONS CONSIDERED BY INDIAN STUDENTS



Q1.

Which career option named here is least considered by Indian students?

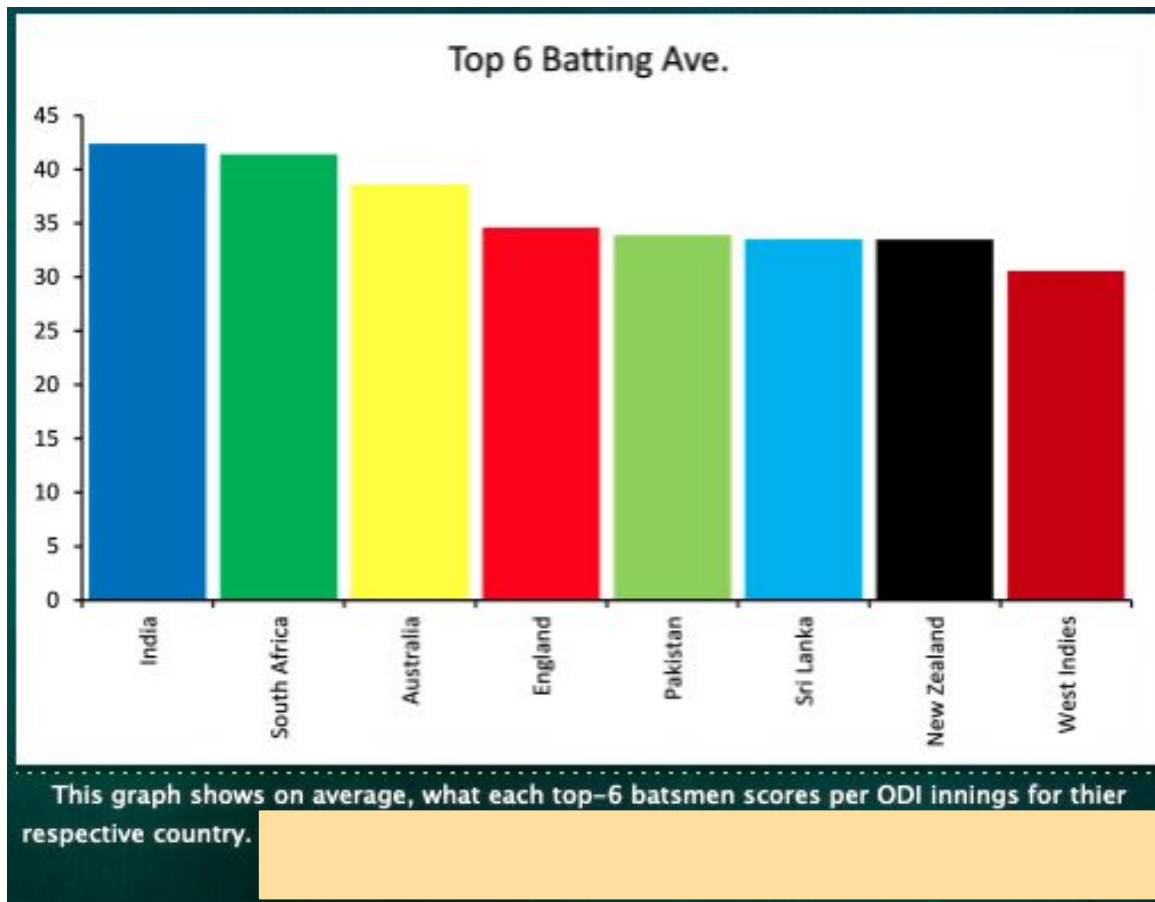
Q2.

Is Design or Management a more preferred choice?

Q3.

Which of the career options are the top 3 considered by students?

Groups 5-8: task 1/2



Q1.

Which country's batsmen score the highest batting average?

Q2.

Which country's batsmen score the lowest batting average?

Q3.

How many countries batsman average score is between 35-30?

Groups 5-8: task 2/2

Debrief

What did you have to pay the most attention to when reading a graph or chart?

Which graph is the most difficult to read? Why?

Today's learning objective- 2

**Analyze and create
graphically represented data**

Keep in mind when making graphics:

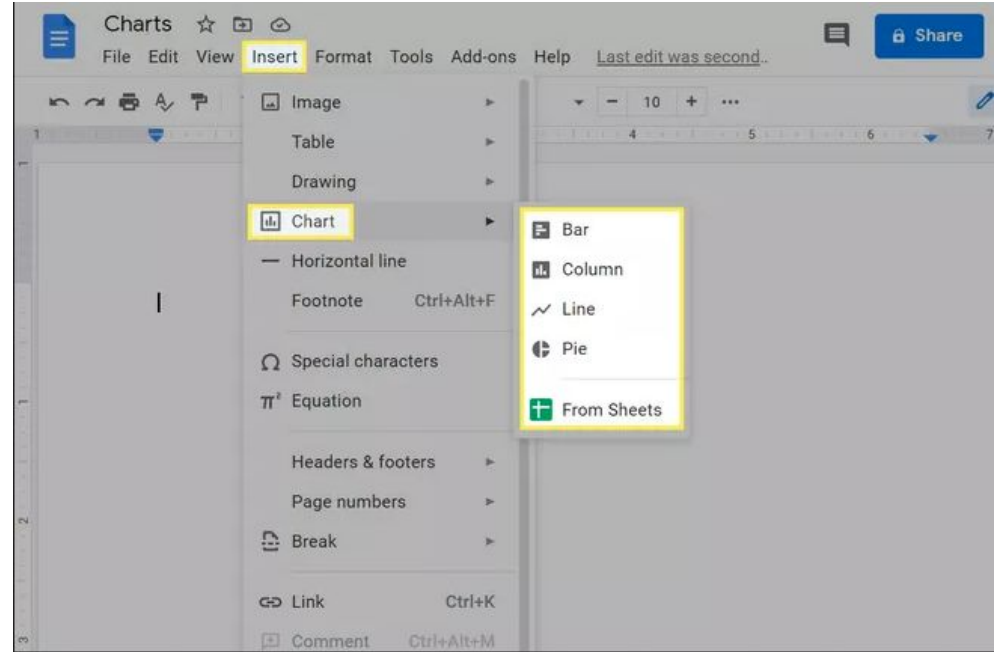
1. Choose the best graphic format for your data (Bar? Pie? Line?).
2. Document your sources.
3. Explain what an abbreviation could be.
4. Number all tables within the text.
5. Ensure all values are consistent across the article.
6. Ensure that formats and headings are consistent.
7. Number and write a caption for your graphic.

(Source:

https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_formatting_and_style_guide/apa_tables_and_figures.html)

Making a Graph on Google Docs

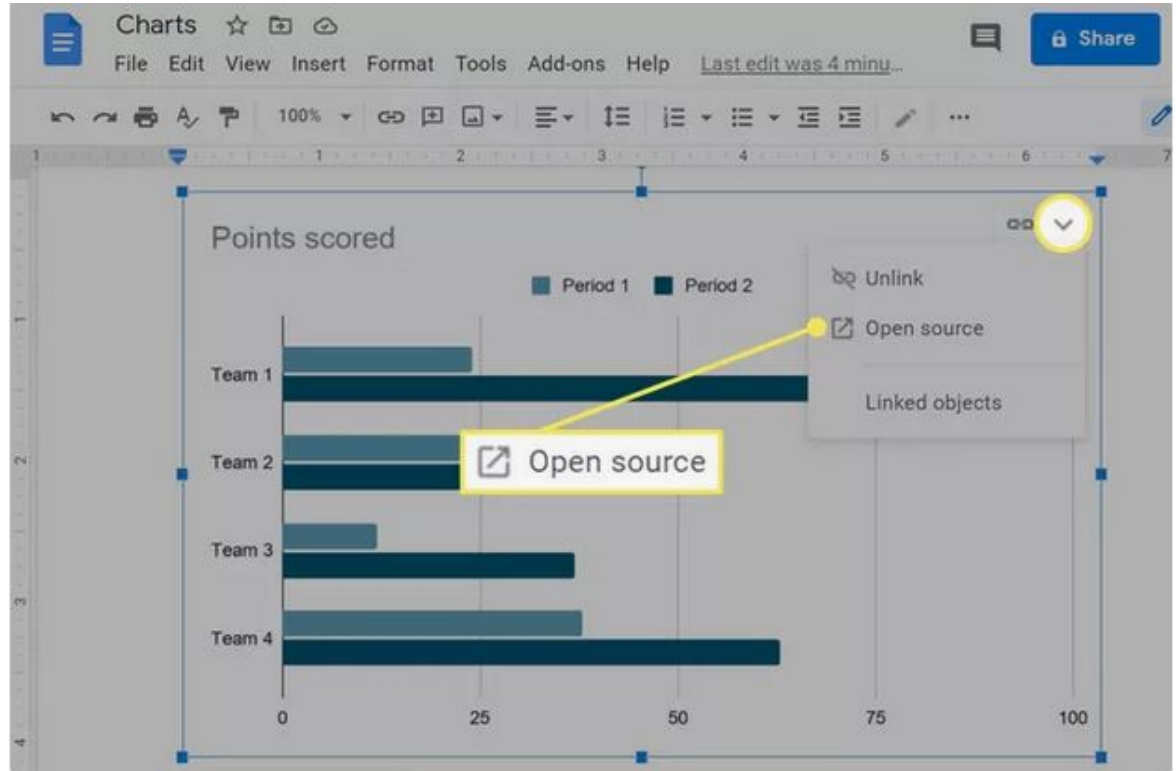
1. Click in the document wherever you want the chart to be.
2. Open the **Insert** menu from the top of the page.
3. From the **Chart** submenu, choose the 'Bar' chart type to have a pre-configured one inserted.
4. You can't just click inside it and start making changes.
5. The real information behind the chart or graph is in **Google Sheets**, where the data is.



Making a Graph on Google Docs

6. Select the chart you want to edit. Use the arrow at the top-right to choose **Open source**.

7. Google Sheets will open. This is where you can edit the chart.

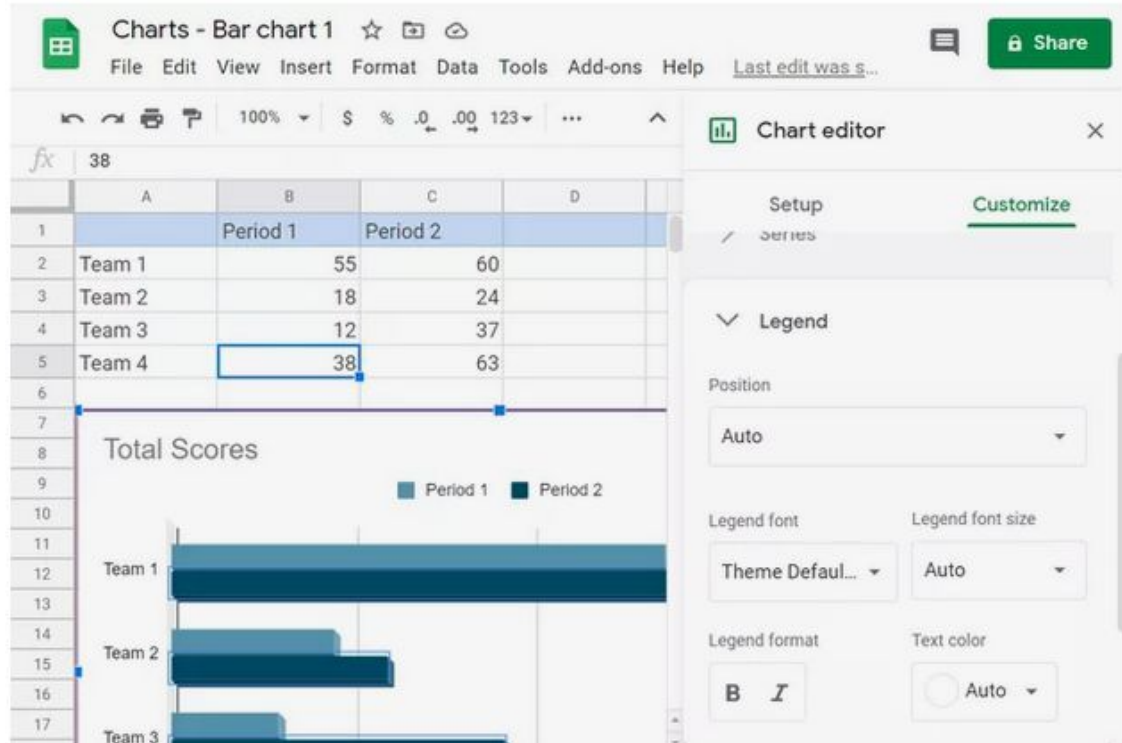


Making a Graph on Google Docs

8. To add or remove data, edit the cells that contain that information.

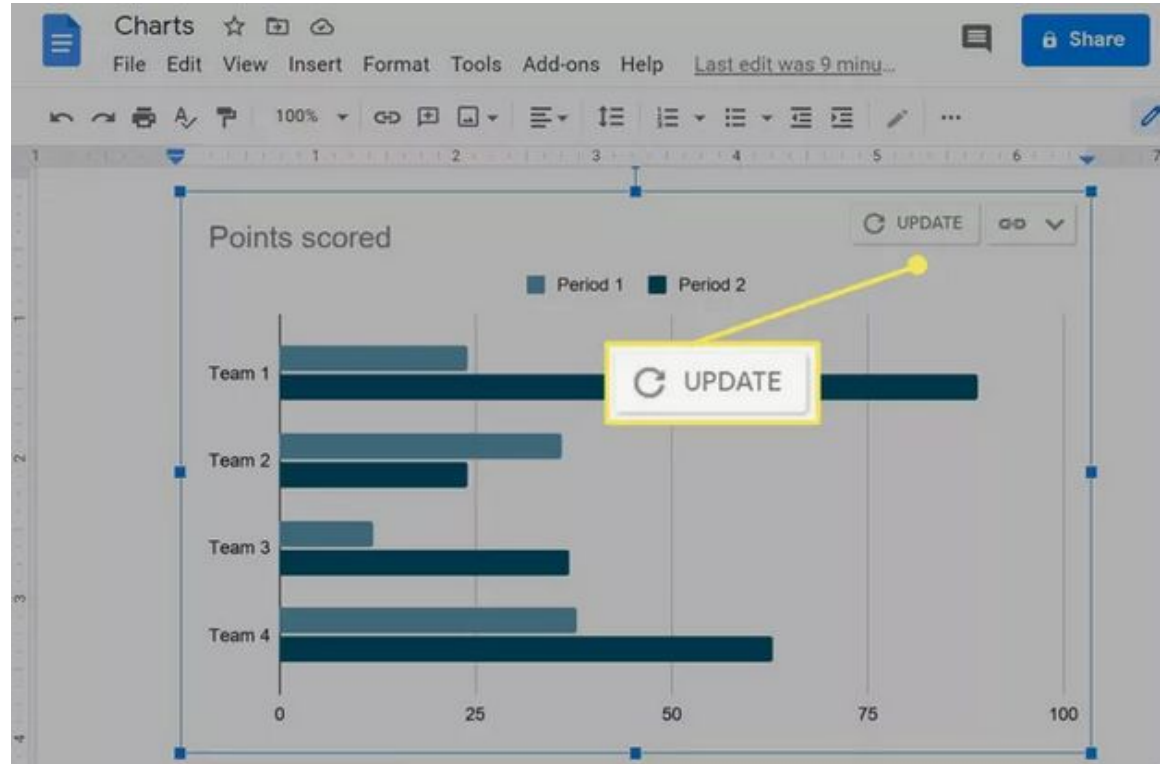
9. Selecting the chart itself and opening its settings is how you modify things like the data range, colors, the legend, axis details, etc.

10. Double-click the chart title to make your own.



Making a Graph on Google Docs

11. Return to Docs when you're done editing and use the **UPDATE** button on the chart to refresh it with any changes you made.



Group activity

1. Choose a table on the page “List of 21st century earthquakes” (https://en.wikipedia.org/wiki/Lists_of_21st-century_earthquakes).
2. Identify the following topics:
 - a. Magnitude
 - b. Location
 - c. Year
 - d. Fatalities/Damage
 - e. Other noticeable topics (if any)
3. Create a table and a graph on a google sheet.



Debrief

In your opinion, which step in creating a table or figure is the most critical step? Why do you think so?